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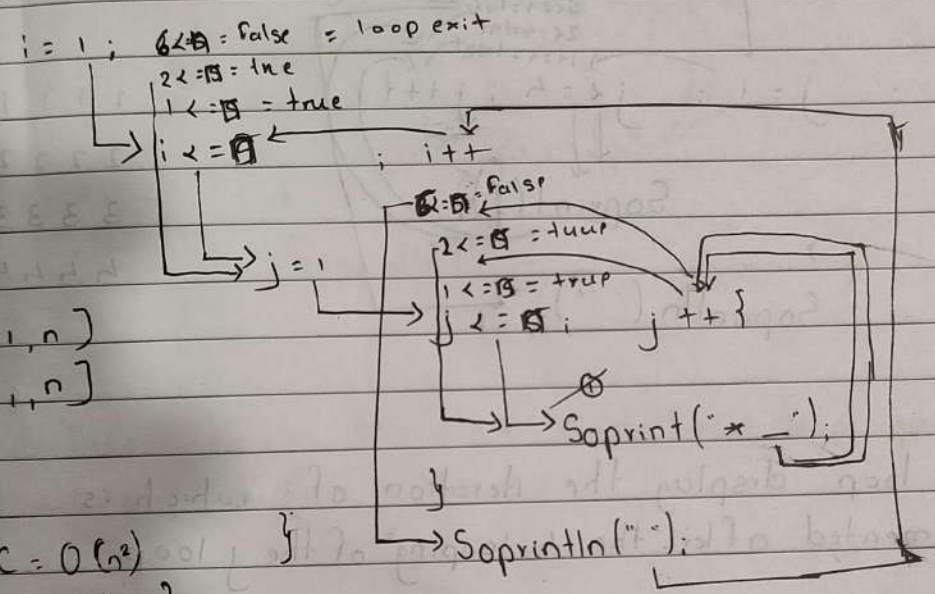
* Assignment - 01

Q.1 take input n from user for

→ n = 5 ; n×n = 5×5

```
* * * *
* * * *
* * * *
* * * *
```

* Dry Run:-




```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

TC = i[1, n]
j[1, n]

* TC = O(n²)
* SC = O(1)

```
import java.util.*;

public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt(); =5

    for(int i = 1; i<=n ; i++){ — n+1

        for(int j = 1; j<=n; j++){ — n×(n+1)

            System.out.print(" * ");— n×n
        }
        System.out.println(" ");
    }
}
```

Inner loop represent no of the columns while outer loop represents the no. rows.

Q2. Print Pattern

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

→ * Dry Run

~~5<=4 = false~~ → loop exit
2<=4 = true
1<=4 = true

```
i = 1 ; i <= 4 ; i++ {
        j = 1 ; j <= 4 ; j++ {
                Soprint(i);
        }
        Soprintln(" ");
}
```

5<=4 = false
2<=4 = true
1<=4 = true

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

* Outer loop display the iteration of i which is incremented after the 4 looping of the j loop

Inner loop display the i of n iteration.

* Code

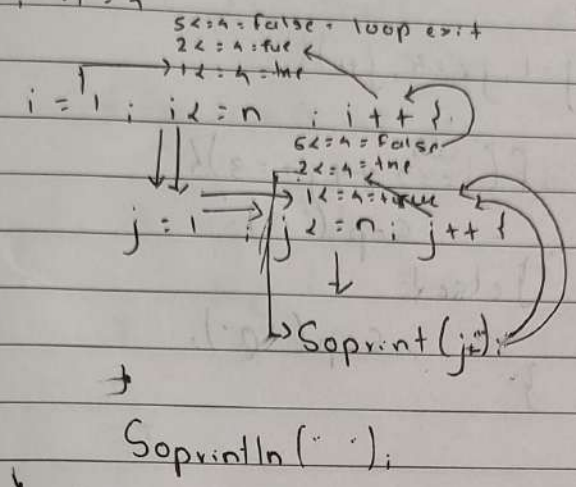
TC : $O(n^2)$
SC : $O(1)$

```
public static void main (String [] args){
        for(int i = 1 ; i <= n ; i++){ — n+1
                for (int j = 1 ; j <= n ; j++){ — n × (n+1)
                        System.out.print(i+" "); — n × n
                }
        System.out.println(" ");
}
```

Q3. Print Pattern , n = 4

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

* Dry Run , n = 4




```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

* Outer loop iterates till n = 4 and Inner loop print the j while loops from 1 to n = 4.

Code.

```
import java.util.*;

    public static void main(String[]args){

        Scanner sc = new Scanner(System.in);
TC - O(n^2)
SC - O(1)       int n = sc.nextInt(); //=4

        for(int i = 1; i<=n; i++){ — n+1

            for(int j = 1; j<=n; j++){ — n(n+1)

                System.out.print(j + " "); n×n
            }
            System.out.println(" ");
        }
    }
```

ex. 3 print

```
o o o o
o o   o
o o o o
```

→ for(int i=1; i<=3; i++){

for(int j=1; j<=4; j++){

if(i==2 || j==3){

sop(" ");

}else{

sop("o");

}

}

}

ex. 4 print

```
o o o o o
o     o
o     o
o o o o o
```

for(int i=1; i<=4; i++){ — n+1

for(int j=1; j<=5; j++){ n*n

if(2<=i<=3 || 2<=j<=4){ TC = O(n²)

SC = O(1)

sop(" ");

}else{

sop("o");

}

}

}

}

$= 1 + (n+1) + n + n \times (3n+2)$

$= 2n + 2 + 3n^2 + 2n$

$= 3n^2 + 4n + 2$

$= 3n^2 + 4n$

$= 3n^2$

$= O(n^2)$

ex. 5

```
    1  2  3  4  5
1   o  o  o  o  o
2   o  o  o
3   o  o
4   o  o
5   o  o
```

for(int i=1; i<=5; i++){

for(int j=1; j<=5; j++){

~~if(i==1 || i==5 ||~~

if(i==j){

sop(0)

}

ex.

```
(1,1) (1,2) (1,3) (1,4) (1,5)
(2,1) (2,2) (2,3) (2,4) (2,5)
(3,1) (3,2) (3,3) (3,4) (3,5)
(4,1) (4,2) (4,3) (4,4) (4,5)
(5,1) (5,2) (5,3) (5,4) (5,5)
```

for(int i=1; i<=5; i++){ — n time

for(int j=1; j<=5; j++){ — n times

TC = O(n²)

SC = O(1)

if(i==j || i+j=6){ n*n = n²

sop(0);

}else{

sop(" ");

}

}

}

Ex.5

```
1 2 3
4 5 6
7 8 9
```

count = 0;

```
for (int i=1 ; i<=3 ; i++){
    for (int j=1 ; j<=3 ; j++){
        count++;
        Sop(count + " ");
    }Soprintln(" ");
}
```

1 2 3

TC = $n \times n = n^2$

TC = $O(n^2)$

SC = $O(1)$

Assignment No: 2

Q1. Hollow Square

* (1,1)	* (1,2)	* (1,3)	* (1,4)	* (1,5)
* (2,1)	o (2,2)	o (2,3)	o (2,4)	* (2,5)
* (3,1)	o (3,2)	o (3,3)	o (3,4)	* (3,5)
* (4,1)	o (4,2)	o (4,3)	o (4,4)	* (4,5)
* (5,1)	* (5,2)	* (5,3)	* (5,4)	* (5,5)

n = 5

Dry Run

```
for ( int i = 1 ; i<=n ; i++){
    for(int j = 1 ; i<=n ; j++){
        if( 2<=j<=4 && 2<=j<=4){
            Soprint(" ");
        }else{
            Soprint("* _ ");
        }
    }
}
```

i<=5
→ j<=5
→ *
j<=5
→ *
j<=5
→ *
j<=5
→ *
j<=5
→ *

i<=5
→ j<=5
→ *
j<=5
→ o
j<=5
→ o
j<=5
→ o
j<=5
→ *

Code.

```
public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    for (int i = 1; i<=n; i++){
        for (int j = 1; j<=n; j++){
            if(2<=i<=4 && 2<=j<=4){
                Soprint(" ");
            }else{
                Soprint("* _ ");
            }
        }Soprintln(" ");
    }
}
```

n×n = n^2

TC = $O(n^2)$

SC = $O(1)$

2.

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

*Dry run n=5 ,

i=1 , i<=n , i++

1<=5
→ 1<=5
→ (i==j || i+j=6)
→ *

2<=5
→ 0

3<=5
→ 0

4<=5
→ 0

5<=5
→ (i==j || i+j==n+1)

```
public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();

    for(int i=1; i<=n; i++){
        for(int j=1; j<=n; j++){

            if(i==j || i+j==n+1){
                System.out.print("* ");
```

```
            }else{
                System.out.print(" ")
            }
        }
    }
}
```

$n \times n = n^2$

TC = $O(n^2)$

SC = $O(1)$

Q3

```
*
* *
* * *
* * * *
* * * * *
```

*Dry run n = 5 ; i<=n ; i++

1<=5
→ i>=1
→ i>=j
→ 1>=1
→ *

2>=1
→ 0

2>=1
→ 2>=1
→ *

2>=2
→ *

2>=3
→ 0

```
public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    for(int i=1; i<=n; i++){
        for(int j=1; j>=i; j++){

            System.out.print("* ");
        }
        System.out.println(" ");
    }
}
```

$n \times n = n^2$

TC = $O(n^2)$

SC = $O(1)$

Q4

```
* * * * *
* * * *
* * *
* *
*
```

* Dry run n=5 ; i=1 ; i<=5 ; i++

i<=5
→ 5>=i ; j--
→ *

4>=1 → *

3>=1 → *

2>=1 → *

1>=1 → *

2<=5

5>=2
→ *

5>=3 → *

3>=2 → *

2>=2 → *

```
public static void main(String[] args){
  for(int i=1; i<=5; i++){
    for(int j=n; j>=i; j--){
      System.out.print("* ");
    }
    System.out.println(" ");
  }
}
```

* n-i+1
5-1+1 = 5
5-2+1 = 4

$n \times n = n^2$
$TC = O(n^2)$
$SC = O(1)$

Q5.

```
*  #  *  #
#  *  #  *
*  #  *  #
#  *  #  *
```

* Dry run i=1 ; i<=n ; i++

1<=4
→ i<=4 ; j++
→ ~~if((i+j)%2==0){~~
System.out.print("*")
→ *

2<=4
→ #

```
for(int i=1; i<=n; i++){
  for(int j=1; j<=n; j++){
    if((i+j)%2==0){
      System.out.print("*");
    }else{
      System.out.print("#");
    }
  }
  System.out.println("");
}
```

$n \times n = n^2$
$TC = O(n^2)$
$SC = O(1)$

ex.6

```
*
* *
* * *
* * * *
* * * * *
```

$n=5$

Dig run

```
for(int i=1 ; i<=5 ; i++){
    for(int j=1 ; j<=5 ; j++){
        if(j>=n-i+1){
            printf("*");
        }else{
            printf(" ");
        }
    }
}
```

$i<=5$
↳ $j<=5$
↳ $j>=n-i+1$
$i<=4$
print(*)

$2<=5$
↳ $4=5-2+1$

$i<=5$
↳ $j<=5$
↳ $j==n-i+1$
$5-1+1$

$TC = O(n^2)$
$SC = O(1)$

ex 7

```
* * * * *
* * * *
* * *
* *
*
```

$i=1$
↳ $j<=5$
↳ $i>=j$
↳

```
if(i>j){
    print(-)
}else{
    (*)
}
```

$TC = O(n^2)$
$SC = O(1)$

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Assignment - 3

Q1. Double Number Pyramid

For $n = 4$

```
1
1 2
1 2 3
1 2 3 4
1 2 3
1 2
1
```

TC = 1, 2, 3 .... n

$\frac{n(n+1)}{2}$

$\frac{4(5)}{2} = 10$

TC = $O(n^2)$

SC = $O(1)$

* Dry Run

$n = 4$

$i = 1$; $i <= n$

↳ $j = 1$; $j <= i$

↳ 1 — 1

1 1 2

1 2 3

1 2 3 4

$i = 2$

↳ $j <= i$

↳ 1 2

$i = 3$

↳ $j <= i$

↳ 3

$i = n-1$; $i >= 1$

↳ $j = 1$; $j <= i$

↳ 1 2 3

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Q2 Number Butterfly

```
1        1
1 2      2 1
1 2 3    3 2 1
1 2 3 4  4 3 2 1
1 2 3 4  4 3 2 1
1 2 3    3 2 1
1 2      2 1
1        1
```

TC = $n^2 + n^2 + n^2 + n^2 = 4n^2$

$n \times n = n^2$

$n \times n = n^2$

$= O(n^2)$

SC = $O(1)$

* Dry Run $n = 4$

$i = 1$; $i <= n$

↳ $j = 1$; $j <= i$

↳ 1

$2 <= n$

$j <= i$

↳ 1 2

* ~~Right~~ Spacing

$j = 1$; $j <= 2 \cdot (n-1)$

↳ . ..

Right

* $j = i$; $j >= 1$

↳ .

$i >= 1$

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Q3 Cross (x) Number Pattern

```
1   1
 2 2
  3
 4 4
5   5
```

$n = 5$

$TC = n \times n = n^2$
$= O(n^2)$

$SC = O(1)$

```
i = 1; i<=n
  ↳ j = 1; j<=n
      ↳ if(j==i || j==n-i+1){
          ↳ " "
      j<=n
        ↳ (else) " 2 " 2
```

Q4. Spiral Layer Pattern

```
3 3 3 3 3
3 2 2 2 3
3 2 1 2 3
3 2   2 3
3 3 3 3 3
```

Dry Run $n = 3$;

```
i = 1 ; i<= 2*n-1
  ↳ j = 1; j<= 2*n-1
      ↳
```

$i = 2n-1$
$j = 2n-1$

$(2n-1) \times (2n-1) = 4n^2 - 4n + 1$

$= 4n^2 - 4n$

$= 4n^2$

$TC = O(n^2)$
$SC = O(1)$

Q5. Pascal triangle

Dry run $n = 5$

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

```
i = 0; i<n; i
  ↳ num 1
  ↳ j = 0; j<=i;
      ↳ n + " "
      n = num * (i-j)
        ↳ 1
j<=i
  ↳ 1 1
```

$TC = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

$TC = O(n^2)$

$SC = O(1)$

Q6 Diamond with continous number.

n = 4

```
   1
  2 3
 4 5 6
7 8 9 10
 11 12 13
  14 15
   16
```

Dry run

m = 4 , num = 1

i = 1 ; i <= n

↳ j = 1 ; j <= i

↳ num

↳ ,

num++

TC = (upper) $1+2+3+\dots+n = \frac{n(n+1)}{2}$

(lower) $(n-1)+(n-2)+\dots+1 = \frac{(n-1)n}{2}$

$TC = \frac{n^2+n}{2} + \frac{n^2-n}{2} = \frac{2n^2}{2} = \boxed{n^2}$

SC = O(1)

i = n-1 ; i >= 1

↳ j = 1 ; j <= i

↳ num

num++

Binary Diamond

```
   1
  0 1
 1 0 1
0 1 0 1
 1 0 1
  0 1
   1
```

Dry Run n = 4

i = 1 ; i <= n

↳ val = i % 2

↳ j = 1 ; j <= i

↳ val + val = 1 - val

1 + 1 = 1 - 1

2 <= n

↳ i <= i

↳ val = 2 % 2 = 0

↳ 0

= 1 % 2 = 1

TC = $O(n^2)$

SC = $O(1)$